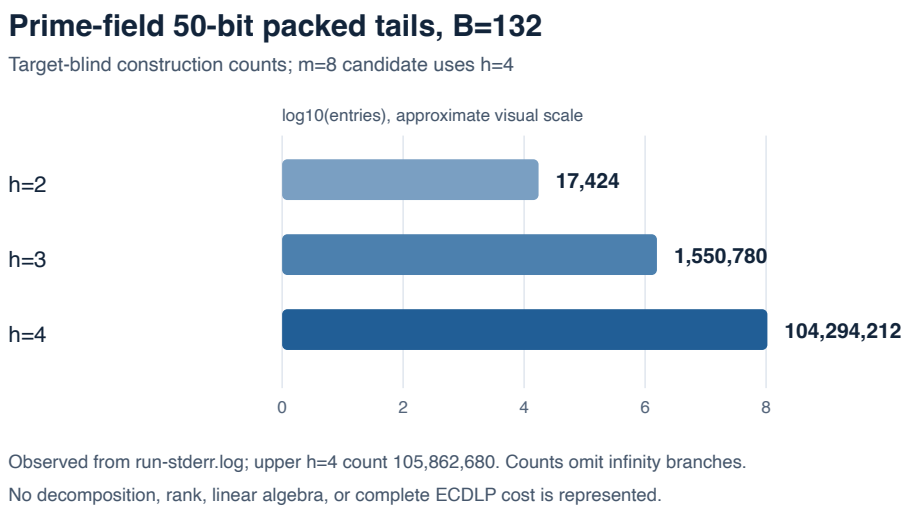
Prime-field 50-bit m=8 packed-tail engineering gate

2026-10-17; TASK-20261017-08bc0f. The bounded target-blind four-point tail construction and independent table check passed. This is an engineering observation, not a complete index-calculus comparison or a speed claim.



Fixed input and arithmetic

The public prime-order curve is over p = 1125899798684033 with a = 779030791575407, b = 827069435865303, and r = 1125899778887903. The fixture SHA256 begins `849ded0bc55a`. `small_x` selected 132 points before any target, in increasing x order, with the smaller y. The point-list SHA256 begins `14c11ea0cde2`. Base construction took 0.005979125 s. The point digest and exact serialization are in `input-manifest.json`. The subgroup order is 50 bits but slightly below the literal value 2^50.

The opt-in Numba producer uses a floating quotient estimate with an exact 64-bit residual correction for p < 2^50. A quotient error below one and a signed residual of magnitude below 2p make the final modular product exact. The C++ `__uint128_t` producer compiled, but fresh native executables stalled before `main` on this host, including two independent hello controls. That host obstruction affects wider fields; it is recorded separately.

Observed construction counts

Tail length	Actual entries	Upper entries	Pair charges
2	17,424	17,556	8,778
3	1,550,780	1,568,336	775,390
4	104,294,212	105,862,680	52,147,172

The derived total is 52,931,340 S3-equivalent pair charges. The sorted h=4 table occupies 1,668,707,392 bytes at 16 bytes per record. The producer recorded 396 inversion batches and 264 exceptional same-x prepends.

Seven small-control tests passed for p=1019 and the same 50-bit curve at h=2,3,4, plus altered-object rejection. The retained 50-bit B8 h4 table contains 2,152 records, and an independent scalar checker replayed all of them with no ordering, bound, or point-sum discrepancy.

Complete-cost boundary

The task envelope was 8 GiB sampled process RSS, 12 GiB scratch, and 14,400 s wall. The wrapper took 982.639 s. Producer phases included 945.218 s for global sort and 18.884 s for final writing; the sort was 96.50% of producer wall. Peak RSS was 5.030 GB (4.684 GiB), and peak scratch was 3.363 GB (3.132 GiB). The table remains in local-only custody under the pinned workspace-volume scratch directory and is excluded from Git.

The independent checker scanned all 104,294,212 records for sort order and bounds with zero failures. It found 101,227,632 distinct x keys and 3,066,580 extra records sharing an x. Its 105 chunk digests and full SHA256 `fa868fc32473...857828b02` match the retained table. Scalar replay passed at 4,096 fixed hash-selected positions and the first and last 256 positions: 4,608 of 4,608 valid. Full scalar replay of every record was not attempted.

This table contains no degree-of-regularity measurement, sparse solver input or runtime, verified relation, independent rank, linear algebra, unseen-target reconstruction, matched rho, or complete end-to-end S. Direct S8 represents seven summands plus the target; eight summands require S9 or a chained formulation. The existing baseline and full-width obstruction remain the comparison references. No canonical ECDLP frontier graph changes from this engineering count chart.